DTC P2503:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

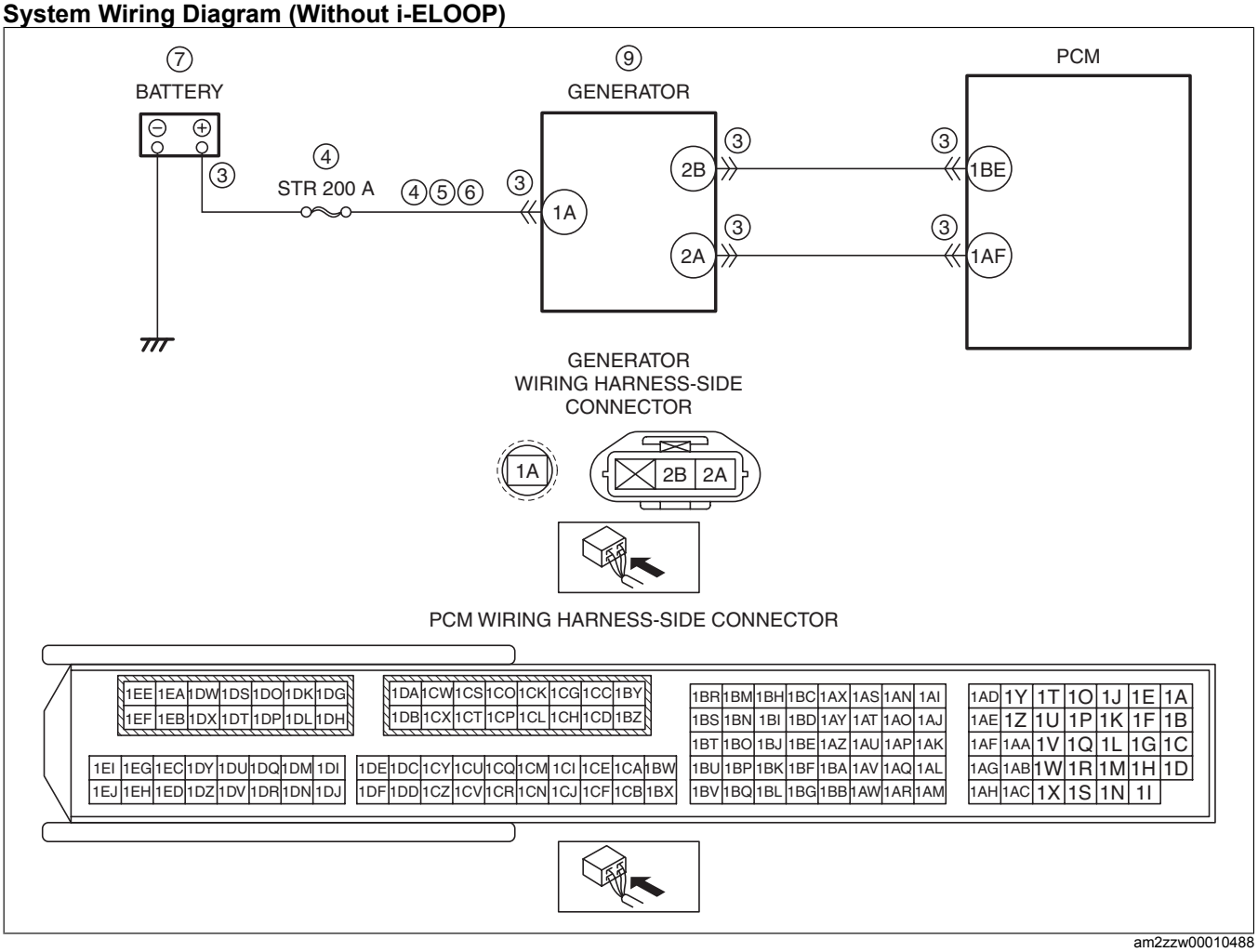
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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs (Without i-ELOOP)

DESCRIPTION	**Generator system: Voltage generated by generator is low**	
DETECTION CONDITION	Determination conditions	• A condition continues for a specified period of time in which the target generator output current calculated by the PCM is **20 A or more** and the generator output voltage is **8.5 V or less**.
	Preconditions	• While engine is running
	Malfunction determination period	• **5 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• PCM • Generator
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Generator control is inhibited.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • The following vehicle conditions differ depending on the type of malfunction: — Vehicle shock may occur due to generator load. — Idling feel due to generator-stop may occur.	
POSSIBLE CAUSE	• Poor connection of the following parts: — Battery — Generator — PCM • Connector or terminal malfunction of the following parts: — Battery — Generator — PCM • STR 200 A fuse malfunction • Short to ground or open circuit in generator charge/discharge circuit — Short to ground in wiring harness between battery positive terminal and generator terminal 1A — Open circuit in wiring harness between battery positive terminal and generator terminal 1A • Drive belt exceeds limit • Battery malfunction • Generator malfunction • PCM malfunction	

System Wiring Diagram (Without i-ELOOP)



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Function Explanation (DTC Detection Outline) (Without i-ELOOP)

- Although a power generation command is made to the generator, a malfunction is diagnosed indicating no output from the generator, and verification of vehicle malfunctions/safety assurance is performed.

Repeatability Verification Procedure (Without i-ELOOP)

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.
3. Leave for **30 s** while idling with high electrical load.

PID Item/Simulation Item Used In Diagnosis (Without i-ELOOP)

- Not applicable

Function Inspection Using M-MDS (Without i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (Without i-ELOOP)
Intention of troubleshooting procedure

- Step 1—6
  - — Perform an inspection of each signal transmission system.
- Step 7
  - — Perform the inspection of the part supplied power to the generator.
- Step 8
  - — Perform a generator drive belt related inspection.
- Step 9
  - — Perform a unit inspection of the generator.
- Step 10—11
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Inspect the connection condition (part installation condition, connector connection condition) for the following parts: — Battery — Generator — PCM • Is the connection condition (part installation condition, connector connection condition) for each part normal?	Yes	Go to the next step.
		No	Connect each part or the connector correctly, then go to Step 10.
3	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Disconnect the connector of the following parts. — Battery — Generator — PCM • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
4	**PURPOSE: INSPECT FUSE** • Remove the STR 200 A fuse. • Inspect the STR 200 A fuse. • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the malfunctioning fuse. If the fuse is damaged: • Replace the fuse. Go to Step 10.
		No	Reinstall the STR 200 A fuse, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**PURPOSE: VERIFY IF SHORT TO GROUND IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator and PCM connectors are disconnected. • Inspect for continuity between generator terminal 1A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 10.
		No	Go to the next step.
6	**PURPOSE: VERIFY IF OPEN CIRCUIT IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, generator and PCM connectors are disconnected. • Inspect for continuity between battery positive terminal (wiring harness-side) and generator terminal 1A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
7	**PURPOSE: VERIFY IF DIAGNOSTIC RESULTS ARE AFFECTED BY INSUFFICIENT EXCITATION CURRENT TO GENERATOR DUE TO POWER SUPPLY PART MALFUNCTION** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery, then go to Step 10. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
8	**PURPOSE: VERIFY IF MALFUNCTION RELATED TO GENERATOR DRIVE BELT AFFECTS DIAGNOSTIC RESULTS** • Inspect the generator drive belt. (See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the indicator mark on the drive belt auto tensioner within the normal range?	Yes	Go to the next step.
		No	Replace the generator drive belt, then go to Step 10. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9	**PURPOSE: DETERMINE INTEGRITY OF GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to the next step. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
		No	Go to the next step.

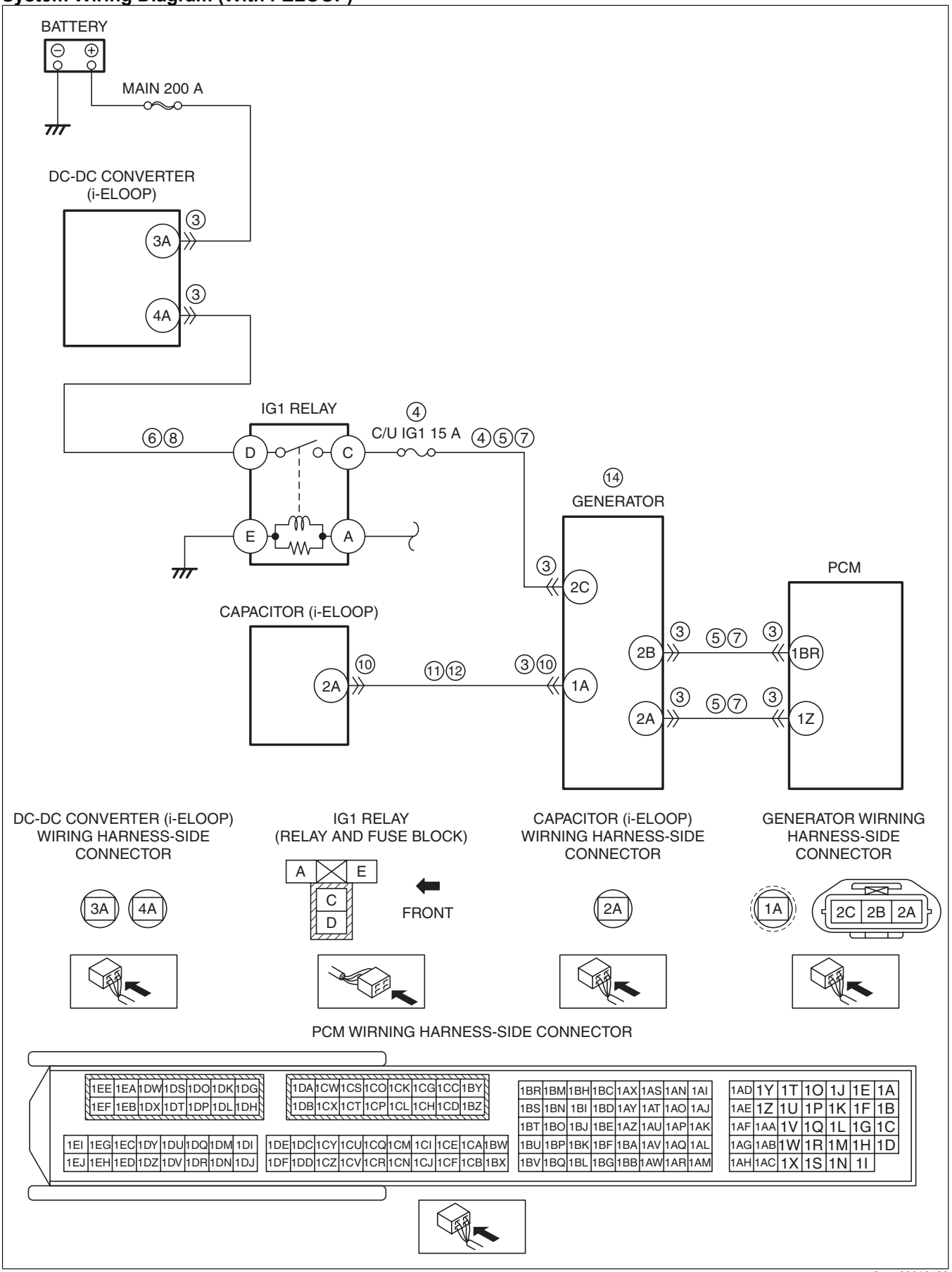
STEP	INSPECTION	RESULTS	ACTION
10	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (Without i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
11	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.

Details On DTCs (With i-ELOOP)

DESCRIPTION	Generator system: Voltage generated by generator is low	
DETECTION CONDITION	Determination conditions	• Any one of the following conditions is met: — Field current demand output to generator is normal range and condition in which field current returned from generator is in non-power generation range continues for specified period — Condition in which malfunction status signal from generator is in returned condition continues for specified period
	Preconditions	• All of the following conditions are met: — While engine is running — Three min have elapsed since pre-charge mode starts — Generator switches to default mode (**14.5 V** constant voltage power generation)
	Malfunction determination period	• **5 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• PCM • Generator
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Generator control is inhibited.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • The following vehicle conditions differ depending on the type of malfunction — A vehicle shock may occur due to generator load — Idling feel due to generator-stop may occur. — i-ELOOP pre-charge mode operates.	

DESCRIPTION	Generator system: Voltage generated by generator is low
POSSIBLE CAUSE	• Poor connection of the following parts: — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Connector or terminal malfunction of the following parts: — Generator — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • C/U IG1 15 A fuse malfunction • Short to ground in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 4A—IG1 relay terminal D — IG1 relay terminal C—Generator terminal 2C — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A — Generator terminal 2B—PCM terminal 1BR — Generator terminal 2A—PCM terminal 1Z • Open circuit in wiring harness between the following terminals: — DC-DC converter (i-ELOOP) terminal 4A—IG1 relay terminal D — IG1 relay terminal C—Generator terminal 2C — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A — Generator terminal 2B—PCM terminal 1BR — Generator terminal 2A—PCM terminal 1Z • Drive belt exceeds limit • Generator malfunction • PCM malfunction

System Wiring Diagram (With i-ELOOP)



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Function Explanation (DTC Detection Outline) (With i-ELOOP)

- The PCM determines that a malfunction occurred based on the condition in which a power generation command is made to the generator but there is no output from the generator, or the generator detected an open/short circuit between the PCM and generator or a power generation command signal abnormality, and verification of vehicle malfunctions/safety assurance is performed.

Repeatability Verification Procedure (With i-ELOOP)

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.
3. Leave for **30 s** while idling with high electrical load.

PID Item/Simulation Item Used In Diagnosis (With i-ELOOP)

- Not applicable

Function Inspection Using M-MDS (With i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**PURPOSE: VERIFY THE PRE-CHARGE MODE CONDITION AND DETERMINE THE MALFUNCTION LOCATION** • Start the engine. • Is the pre-charge warning displayed in the display?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
4	**PURPOSE: VERIFY IF GENERATOR OUTPUT IS AFFECTED BY DTC OCCURRING FROM DC-DC CONVERTER (i-ELOOP) RELATED PART** • Switch the ignition off, then ON (engine off). • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present? **Note** • If DTC P1628:11 is stored, perform the on-board diagnostic function first from DTC P1628:11.	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].) Go to the next step.
		No	Go to the next step.
5	**PURPOSE: VERIFY DTC** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (With i-ELOOP)
Intention of troubleshooting procedure

- Step 1
  - — Perform inspection of generator voltage.
- Step 2—12

— Perform inspection of each signal transmission system.
— Perform the inspection of the part supplied power to the generator.

- Step 13
  — Perform a generator drive belt related inspection.
- Step 14
  — Perform a unit inspection of the generator.
- Step 15—17
  — Verify that primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: DETERMINE THE MALFUNCTION CAUSE BY VERIFYING THE OUTPUT VOLTAGE OF GENERATOR TERMINAL 1A.** • Start the engine. • Measure the voltage at the generator terminal 1A (wiring harness-side). • Is the voltage **14—15 V**?	Yes	Go to the next step.
		No	If the voltage is **approx. 0 V**: • Go to Step 9. If the voltage is except for **approx. 0 V**: • Go to the next step.
2	**PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Remove the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Inspect the connection condition (part installation condition, connector connection condition) for the following parts: — Generator — PCM — DC-DC converter (i-ELOOP) • Is the connection condition (part installation condition, connector connection condition) for each part normal?	Yes	Go to the next step.
		No	Connect each part or the connector correctly, then go to the next step.
3	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Disconnect the connector of the following parts. — Generator — PCM — DC-DC converter (i-ELOOP) • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.
4	**PURPOSE: INSPECT FUSE** • Remove the C/U IG1 15 A fuse. • Inspect the C/U IG1 15 A fuse. • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and generator terminal 2C. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the fuse. If the fuse is damaged: • Replace the fuse. Go to the next step.
		No	Reinstall the C/U IG1 15 A fuse, then go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the generator, PCM and DC-DC converter (i-ELOOP) connectors are disconnected. • Remove the IG1 relay. (See RELAY LOCATION.) • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — IG1 relay terminal C — Generator terminal 2B — Generator terminal 2A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay terminal C—Generator terminal 2C • Generator terminal 2B—PCM terminal 1BR • Generator terminal 2A—PCM terminal 1Z **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Go to the next step.
6	**PURPOSE: INSPECT IG1 RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND** • Verify that the generator, PCM and DC-DC converter (i-ELOOP) connectors are disconnected. • IG1 relay is removed. • Inspect for continuity between IG1 relay terminal D (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and IG1 relay terminal D. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Go to the next step.
		No	Go to the next step.
7	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the generator, PCM and DC-DC converter (i-ELOOP) connectors are disconnected. • IG1 relay is removed. • Inspect for continuity between the following terminals (wiring harness-side): — IG1 relay terminal C—Generator terminal 2C — Generator terminal 2B—PCM terminal 1BR — Generator terminal 2A—PCM terminal 1Z • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay terminal C—Generator terminal 2C • Generator terminal 2B—PCM terminal 1BR • Generator terminal 2A—PCM terminal 1Z **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
8	**PURPOSE: INSPECT IG1 RELAY POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Verify that the generator, PCM and DC-DC converter (i-ELOOP) connectors are disconnected. • IG1 relay is removed. • Inspect for continuity between DC-DC converter (i-ELOOP) terminal 4A (wiring harness-side) and IG1 relay terminal D (wiring harness-side). • Is there continuity?	Yes	Go to Step 15.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and IG1 relay terminal D. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Go to Step 13.
9	**PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Inspect the connection condition (part installation condition, connector connection condition) for the following parts: — Generator terminal 1A — Capacitor (i-ELOOP) • Is the connection condition (part installation condition, connector connection condition) for each part normal?	Yes	Go to the next step.
		No	Connect each part or the connector correctly, then go to the next step.
10	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Disconnect the connector of the following parts. — Generator terminal 1A — Capacitor (i-ELOOP) • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.
11	**PURPOSE: VERIFY IF SHORT TO GROUND IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the capacitor (i-ELOOP) and generator terminal 1A connectors are disconnected. • Inspect for continuity between generator terminal 1A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between capacitor (i-ELOOP) terminal 2A and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
12	**PURPOSE: VERIFY IF OPEN CIRCUIT IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS** • Verify that the capacitor (i-ELOOP) and generator terminal 1A connectors are disconnected. • Inspect for continuity between capacitor (i-ELOOP) terminal 2A (wiring harness-side) and generator terminal 1A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between capacitor (i-ELOOP) terminal 2A and generator terminal 1A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
13	**PURPOSE: VERIFY IF MALFUNCTION RELATED TO GENERATOR DRIVE BELT AFFECTS DIAGNOSTIC RESULTS** • Inspect the generator drive belt. (See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the indicator mark on the drive belt auto tensioner within the normal range?	Yes	Go to the next step.
		No	Replace the generator drive belt, then go to the next step. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
14	**PURPOSE: DETERMINE INTEGRITY OF GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to the next step. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
		No	Go to the next step.
15	**PURPOSE: VERIFY CONDITIONS OF BATTERY** • Inspect the battery. (See BATTERY INSPECTION.)	—	Follow the inspection instructions, then go to the next step.
16	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (With i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
17	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.